

Introduction

Understanding the physics of so-called ‘strange metals’ is a central challenge of modern condensed matter physics. One of the most notable properties of the high temperature cuprate superconductors, known already for thirty-five years, is a d.c. resistivity ρ that depends linearly on temperature (T) from just above the superconducting transition temperature (T_c) to hundreds of kelvin, in one notable case crossing two decades of temperature [1]. In any single cuprate material, $\rho \sim \rho_0 + AT$ (with ρ_0 the residual resistivity) when it is tuned to be close to its maximum T_c , suggesting a link between the T -linear resistivity and the high temperature superconductivity [2]. Later, it was realized that not only the power law, but also its prefactor A are significant. Optical conductivity data on cuprates showing T -linear resistivity provided evidence of a resistive scattering rate $\frac{1}{\tau_{tr}}$ varying as $k_B T / \hbar$, a rate postulated to be the highest allowed in a strongly interacting quantum system and termed ‘Planckian dissipation’ [3–5]. Interest in T -linear resistivity in strongly correlated electron systems grew with the discovery of its existence in other systems tuned to the proximity of quantum critical points, including heavy fermion materials, iron-based and organic superconductors and transition metal oxides [6–10]. Optical conductivity data is sparser in these systems, but analysis of d.c. resistivity data in certain systems for which there was experimental information on the Fermi surfaces and effective masses m^* concluded that the observed values of A are such that $\frac{1}{\tau_{tr}} = \alpha \frac{k_B T}{\hbar}$ with $0.7 < \alpha < 2.7$ [10]. Similar analyses applied more recently to d.c. transport data from cuprates [11] and twisted bilayer graphene [12] reached similar conclusions, while T -linear resistivity was shown to persist to temperatures as low as 10 mK in YbRh₂Si₂ [13].

The analyses described above rely on accurate knowledge of the quasiparticle parameters of the materials in question, and the use of different Fermi surface averages of the Fermi velocity v_F for $\rho \propto \frac{1}{v_F(k)}$ and $m^* \propto \overline{v_F(k)}$ [14]. This issue can be mitigated by having information on all the Fermi surface sheets, but the different ways of averaging within individual sheets can introduce error if v_F is strongly k -dependent. A second important issue when considering Planckian dissipation is the origin of the scattering that produces it. The Planckian rate $\frac{1}{\tau_{tr}} \cong$

$\frac{k_B T}{\hbar}$ has been noted at high temperatures in electron-phonon systems since the time of Peierls [15], [16] and understood in terms of a temperature-dependent scattering cross-section for quasi-elastic processes. The fact that it is approximately bounded in real materials has been postulated to be because of a bound on the stability of the metallic state to polaron formation, i.e. a limit on how high electron-phonon coupling can be while still leading to a conventional metal [17]. In contrast, electron-electron scattering is thought to give $\frac{1}{\tau_{tr}} \cong \frac{k_B T}{\hbar}$ because of inelastic processes [18] that are more naturally linked to the concept of an equilibration time [14]. The issue of the temperature range over which electron-electron or electron-phonon scattering are microscopically responsible for the observed $\rho \sim \rho_0 + AT$ is a matter of active debate in the literature [19]. There is also surprisingly little systematic investigation of the level to which, when there is scattering at the Planckian rate, A and ρ_0 are independent.

The material on which this paper is based, PdCrO_2 , is ideal for careful experimental investigation of the issues outlined above. It consists of alternating layers of highly conducting Pd, coupled to Mott insulating layers of CrO_2 in which Cr^{3+} is thought to be in the high-spin $3d^3$ configuration with an effective moment corresponding to spin-3/2 [20–22]. Its Weiss temperature (θ_W) is approximately 500 K but magnetic order sets in only at $T_N = 38$ K. The ratio between its out-of-plane resistivity ρ_c and in-plane resistivity ρ_{ab} is larger than 150 [23], justifying a 2D approximation when analyzing its properties. Strikingly, ρ_{ab} is T -linear from approximately 150 K to at least 500 K [24].

PdCrO_2 is particularly well-suited to the analysis of this T -linear in-plane resistivity for a number of reasons. Consistent with the large value of ρ_c/ρ_{ab} , the high temperature Fermi surface is a simple cylinder of nearly hexagonal cross-section, with a nearly constant $v_F(\mathbf{k})$ in a 2D Brillouin zone [25–27]. This high temperature Fermi surface is almost identical to that of its partner material PdCoO_2 in which Co^{3+} is in the non-magnetic low-spin $3d^6$ configuration, enabling a quantitative measurement of the degree of magnetic scattering in PdCrO_2 by empirical comparison between the properties of the two materials. For both materials, the residual resistivity is tiny in as-grown crystals, providing a well-controlled starting point for data interpretation. Finally, the physics of PdCrO_2 has been shown to be well described by a

low energy Kondo lattice Hamiltonian describing the inter-layer coupling of conduction electrons with the local moments in the Mott insulator layer [27], providing firm foundations for future theoretical work. This combination of properties makes PdCrO₂ attractive as one of the materials best suited for analysis of T -linear metallic resistivity.

Working with single crystals and cutting micro-devices from them to perform electrical resistivity measurements, we show that $\frac{1}{\tau_{tr}} = 0.9 \frac{k_B T}{\hbar}$ in as-grown crystals, and remains unchanged as ρ_0 is increased by a factor of twenty by the deliberate introduction of defects by high-energy electron irradiation. We analyse heat capacity and thermal conductivity data and discuss the origin of the dominant scattering mechanism leading to the observed Planckian resistivity, and comment on the role that Kondo lattice physics might play in accounting for our observations.

Experiment

Sample growth

Single crystals of PdCoO₂ and PdCrO₂ were grown in quartz tubes via methods discussed in ref. [28] and ref. [29]. For the microstructuring described below we selected platelets $\sim 5\text{--}20$ μm thick, with lateral dimensions and $\sim 300\text{--}700$ μm .

Electron irradiation

Crucially for the analysis, the scattering centres introduced to the crystal should be point-like defects and no large voids or columnar defects should be created. High-energy electron irradiation is the ideal technique to achieve this type of disorder. The collision kinetics of 2.5 MeV electrons with much heavier atoms allow transmission of enough energy to an atom to displace it from its lattice site, but not enough for the displaced atom to create a significant number of additional defects. The collisions each therefore create an individual vacancy plus an interstitial atom, known as a Frenkel pair. The 2.5 MeV electrons have a large penetration range estimated to be 1.8 mm in delafossite metals, so the probability of any electron

undergoing more than one collision in samples a few tens of microns thick is negligible. Further details can be found in Ref. [30].

The 2.5 MeV electron irradiation was performed at the SIRIUS Pelletron linear accelerator operated by the Laboratoire des Solides Irradiés (LSI) at the École Polytechnique in Palaiseau, France. During the irradiation, the sample is immersed in a bath of liquid hydrogen at a temperature of ~ 22 K, and we are able to perform an in-situ four-point resistance measurement by pausing at regular intervals and therefore monitor the increase of resistivity as a function of electron dose.

Sample preparation

In order to enable a reliable low-temperature resistivity measurement of delafossites, we used focused ion beam sculpting to increase the signal-to-noise ratio and decrease the geometrical uncertainties. The standard method of sample mounting for microstructuring requires a layer of glue to attach the crystal to a substrate, as was done in previous transport studies [31]. However, glue degrades quickly in the electron beam, motivating us to use the 'epoxy-free' method of mounting PdCoO_2 described in [30]. A scanning electron micrograph of a PdCoO_2 device structured using this method is shown in fig. 1a. The crystal is held on a substrate by electrostatic forces and pinned by deposited Pt contacts providing a good mechanical connection between the crystal and gold-coated substrate. The second sputtered 150-nm gold layer ensures contact resistance on the order of 1Ω . Unfortunately, the 'epoxy-free' method does not apply to PdCrO_2 as a strong mechanical coupling between the substrate and the crystal breaks the device during thermal cycling. To decouple the crystal from the substrate, we mounted it in two-component glue and sputtered a 150-nm gold layer. The deposited Pt leads play the role of a bridge connecting PdCrO_2 with the substrate. The glue was removed underneath the sample by oxygen plasma etching, leaving crystal 'hanging' on Pt leads, as shown in fig. 1b. We used a focused ion beam (FIB) to sculpt the PdCrO_2 for resistivity measurements as the final step.

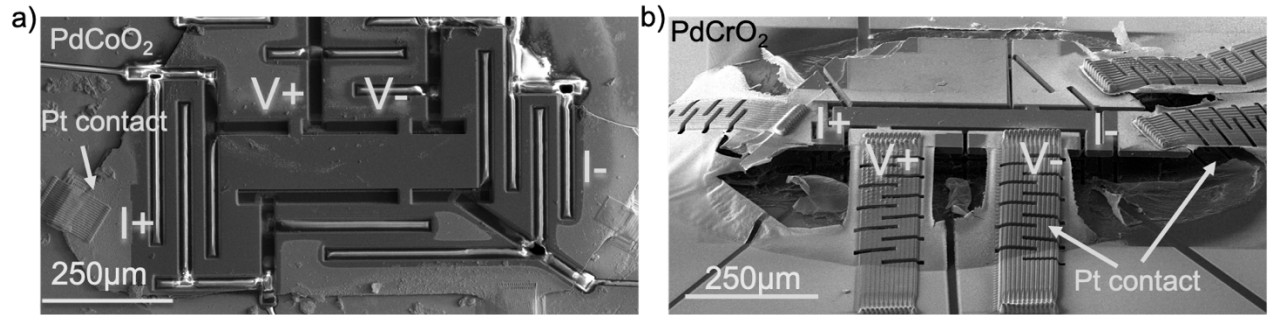


Figure 1

- (a) SEM image of a microstructured PdCoO₂ sample mounted by 'epoxy-free' method. (b) SEM image of the PdCrO₂ device mounted using 'free-standing' method. The measured regions of two devices appear to be darker in the SEM image because the gold is removed from it. The rest of the devices, Pt contacts, and substrate are covered with sputtered gold. The Pt contacts are slotted to decrease their spring constant and give the overall device enough mechanical flexibility to avoid the PdCrO₂ fracturing during cool-downs.

Thermal conductivity **to follow**

Results

In Fig. 2 we show the resistivity calculated for the PdCrO_2 microstructure shown in Fig. 1, for which the geometrical factors are well-defined, highlighted against previous work going to higher temperatures for which data were taken on bulk single crystals [24]. The agreement is seen to be excellent, giving confidence that the T -linear resistivity seen between 150 K and 300 K in the PdCrO_2 microstructure is representative of data going to much higher temperatures of over 500 K. The gradient of the T -linear term is $0.025 \mu\Omega\text{cm/K}$. As in all previous reports, there is a sharp drop in resistivity below T_N .

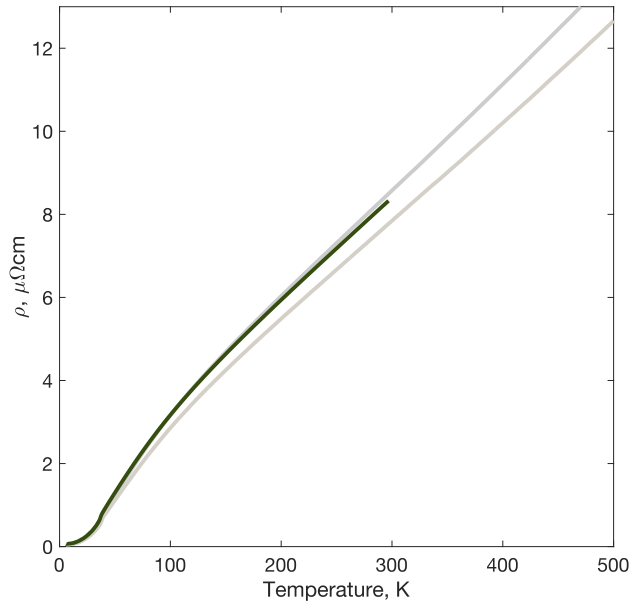


Figure 2: In-plane resistivity data for the PdCrO_2 microstructure shown in Fig. 1 (green) compared with data from previous work [24] on two single crystals (grey). The difference between the results from the two crystals is due to inevitable uncertainty in the geometrical factors used to convert resistance to resistivity when working with traditional hand-mounted silver epoxy contacts. Geometrical errors can be made much smaller with microstructures. Overall, the agreement between the three measurements is excellent. A T -linear resistivity is observed between approximately 150 K and the highest measurement temperature of 500 K.

De Haas-van Alphen measurements performed on PdCrO_2 show that its high temperature Fermi surface (sampled at low temperatures thanks to magnetic breakdown across small gaps

in the low temperature antiferromagnetically ordered state) has an average cyclotron mass $m^* = 1.5 m_e$, where m_e is the electron mass [24], [32]. This corresponds to a Fermi-surface averaged Fermi velocity of approximately $7.5 \times 10^5 \text{ ms}^{-1}$. In common with observation and calculation on other delafossite Fermi surfaces [31], [33–35], angle-resolved photoemission spectroscopy [25–27] shows that the variation of the Fermi velocity about this mean is only $\pm 7\%$. Calculating the carrier density n from the Fermi volume of the half-filled conduction band and using $m^* = 1.5 m_e$ gives $\frac{1}{\tau_{tr}} = \frac{ne^2\rho}{m^*} = 0.9 \frac{k_B T}{\hbar}$. The favourable Fermi surface parameters summarized above mean that the degree of averaging required for the analysis is very small. This is one of the most reliable possible of all such estimates of $\frac{1}{\tau_{tr}}$, so the closeness of the result to the Planckian value is particularly striking.

Our choice of microstructures for the current experiment was motivated by the desire to perform a careful study of the effect of changing the elastic scattering rate of PdCrO_2 by adding disorder. Although this can in principle be done by either irradiating or chemically doping different samples [30], [36–38], the inevitable uncertainties in geometrical factors between those samples is difficult to separate with confidence from the intrinsic effects of disorder. Our goal was to obtain a precise measurement of the effect of the extra elastic scattering on the T -linear term in ρ_{ab} , so it was vital to eliminate geometrical uncertainty from the experiment. As described above, the high energy electron irradiation that provides the required point defects places the sample in a harsh environment, so even working with one single crystal mounted with traditional silver paint contacts is risky. If one of those contacts moved or had to be repaired, the change in geometrical factor could ruin the experiment. In a microstructure of the kind shown in Fig. 2, the geometrical factors are determined by the sculpting of the sample, so as long as it does not break during the experiment, even the contacts between the sample and the measurement wires can be repaired, if necessary, without any change to the geometry of the measurement.

The increase of the resistivity of PdCoO_2 , and PdCrO_2 as a function of electron dose is shown in fig. 4a. In these two compounds, the dependence of resistivity on dose is linear and has the same slope in the investigated range. The extra resistivity is dominated by defects in the conductive Pd planes, as expected in such two-dimensional systems [30].

The results of two long-time irradiations of the same PdCrO₂ microstructure are shown in Fig. 4b. In each case the sample was left overnight in the electron beam, and its resistivity measured after transporting it from Paris to Dresden. The second irradiation took place in a different beam time to the first one, so overall the experiment took six months, but it produced a conclusive result: within experimental error the gradient of the T -linear resistivity is unchanged in spite of increasing the residual resistivity by over a factor of twenty, from 40 to 900 nΩcm.

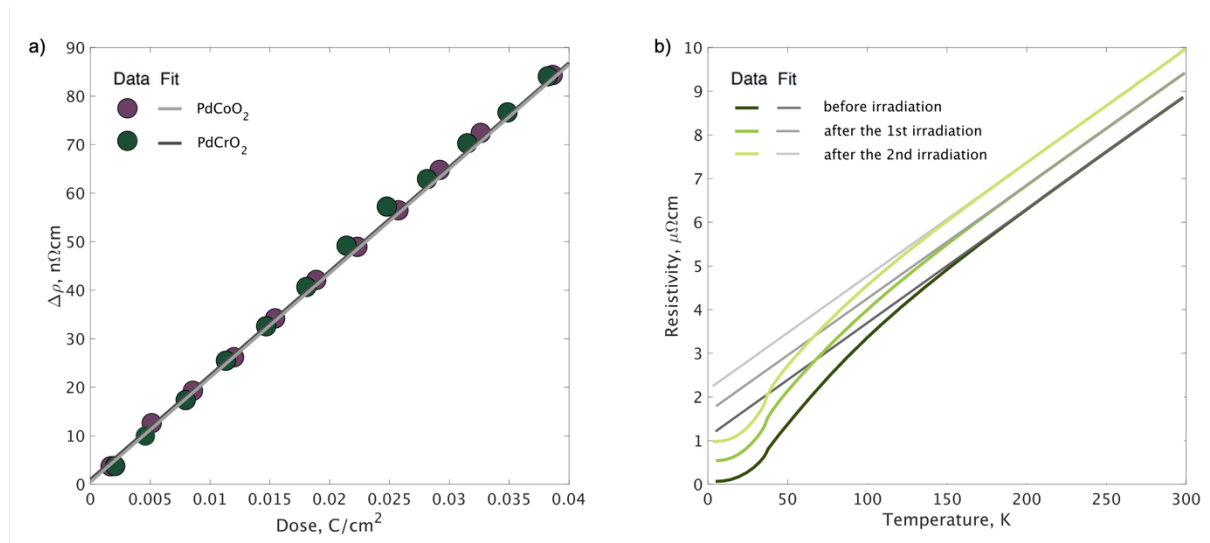


Figure 3:

(a) Increase of resistivity as a function of electron dose for two delafossite metals: PdCoO₂, and PdCrO₂.
(b) change to the in-plane resistivity of the PdCrO₂ microstructure shown in Fig. 1, in its as-prepared state and then after two successive exposures to high energy electron radiation with defect concentrations of 0.16% and 0.29%. The gradient of the T -linear resistivity is unchanged, within small experimental errors, although the residual resistivity is increased by over a factor of twenty.

To summarize the conclusions that can be drawn from analysis of Figs. 2 and 3: the scattering rate in PdCrO₂ can be extracted from the d.c. resistivity with an unusually high degree of confidence because of its simple Fermi surface and favourable quasiparticle parameters. It is 90% of the Planckian value, and remains unchanged within experimental error even when significant levels of disorder are introduced, substantially changing the levels of elastic disorder scattering. These are two of the key experimental observations that we report. For

the remainder of the paper we consider the scattering mechanisms that could lead to the observed behaviour.

First, we examine the thermal conductivity κ , for which data are presented in Fig. 4a. As expected of a high-conductivity metal, it is large, reaching nearly 300 W/Km at its peak value. It rises rapidly below T_N , qualitatively consistent with the behaviour of the electrical conductivity. Quantitative comparison of electrical and thermal conductivity comes from calculating the Lorenz ratio $L = \frac{\kappa\rho}{T}$. In systems for which electrons dominate heat conduction, and the same average scattering rate determines both the thermal and electrical conductivity, $L = L_0 = \frac{\pi^2}{3} \left(\frac{k_B}{e} \right)^2$. The condition for this is that the scattering be quasi-elastic, i.e. that the average energy relaxation due to the scattering is much smaller than the average momentum relaxation. This is satisfied in any metal at sufficiently low temperatures in the regime in which impurity scattering dominates.

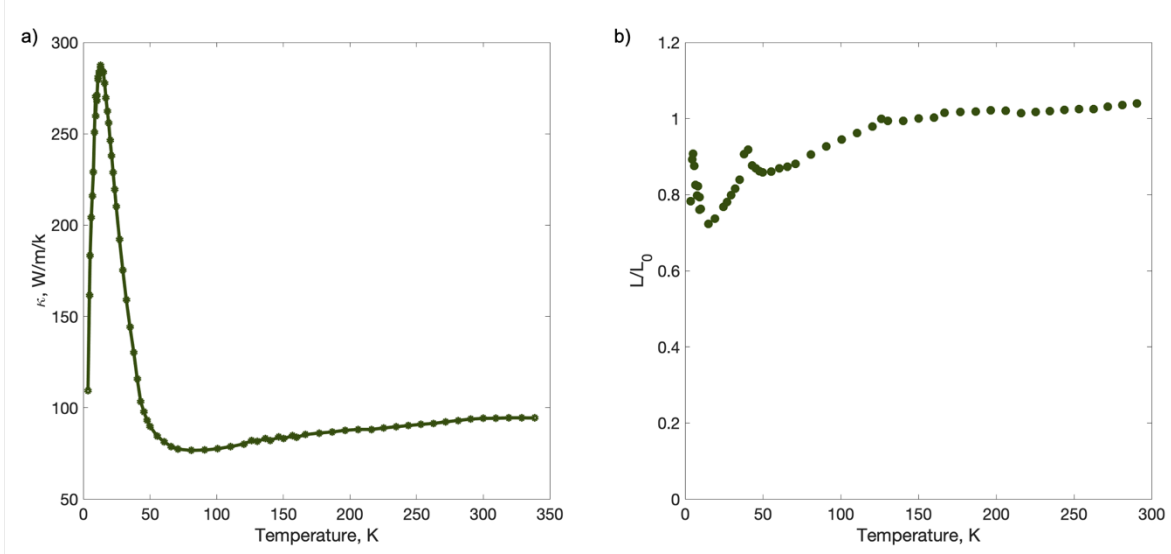


Figure 4: a) Thermal conductivity and b) Lorenz ratio of PdCrO₂ as a function of temperature

In PdCrO₂, L is seen to be within 5% of L_0 for all measured temperatures above 100 K. Similar behaviour is seen in conventional metals, and attributed to electron-phonon scattering. In PdCrO₂, however, a body of evidence points to a more complex situation. Firstly, the sharp drop of resistivity below T_N hints at rather strong magnetic scattering which is frozen out as the magnetically ordered state is entered. The existence of the non-magnetic sister

compound PdCoO_2 gives the opportunity for a complementary estimate of the strength of electron-phonon scattering in PdCrO_2 . The two compounds have very similar total atomic mass, so their phonon spectra are therefore also expected to be similar, as confirmed by direct observation of similar optic phonon frequencies in the two compounds [39], [40]. To a first approximation, then, the properties of PdCoO_2 can be used to estimate the contribution of phonons to the properties of PdCrO_2 . We first consider the entropy of the two materials. In Fig. 5 a) we show heat capacity (C_P) measurements from 4 K to 390 K. For both PdCrO_2 and non-magnetic PdCoO_2 , C_P is still rising quite rapidly at 390 K, and in PdCrO_2 has reached only 85% of the classical Dulong-Petit value, consistent with previous estimates of a high Debye temperature of approximately 500 K [39]. More striking still is the difference in heat capacity between magnetic PdCrO_2 and non-magnetic PdCoO_2 , which remains significant to our highest temperature of measurement. In Fig. 5 b) we show the calculated entropy of the two materials, as well as the difference between the two. In the simplest assumption, this difference entropy could be classified as the magnetic contribution. By 390 K, it is much closer to saturation than the lattice entropy of PdCoO_2 , but with a value 30% higher than the $R \ln 4$ that would be the naïve expectation for freely fluctuating local spin 3/2 moments.

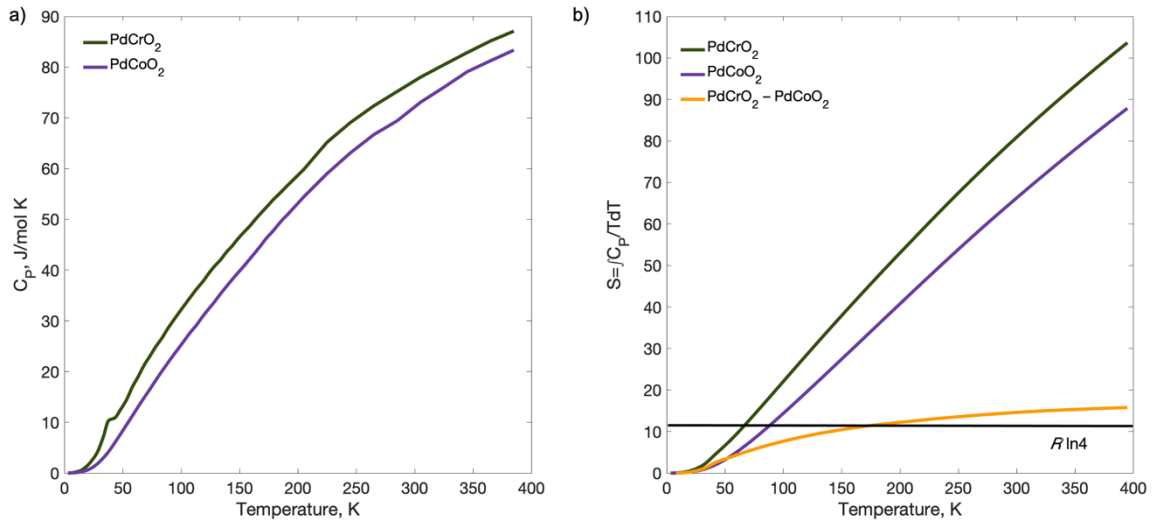


Figure 5: a) The total heat capacity of magnetic PdCrO_2 and non-magnetic PdCoO_2 as a function of temperature up to 390 K. b) The entropy of the two compounds over the same range of temperatures, and the difference entropy, which in a naïve interpretation would be attributed to the magnetic contribution.

The empirical comparison of the resistivities of the two compounds provides further evidence that the spin degrees of freedom scatter the conduction electrons strongly. In Fig. 6 we show the resistivity of the PdCrO₂ and PdCoO₂ microstructures studied in this project, overlaying the previously-reported results on bulk single crystals from Ref. [24]. The difference between the two, also plotted in Fig. 6, is substantial – larger than the resistivity of PdCoO₂. Taken together with the behaviour of the resistivity of PdCrO₂ across its magnetic transition, the natural explanation is strong scattering of the conduction electrons from the Pd layers from the spins in the CrO₂ layers, due to the rather strong coupling between the two subsystems [27].

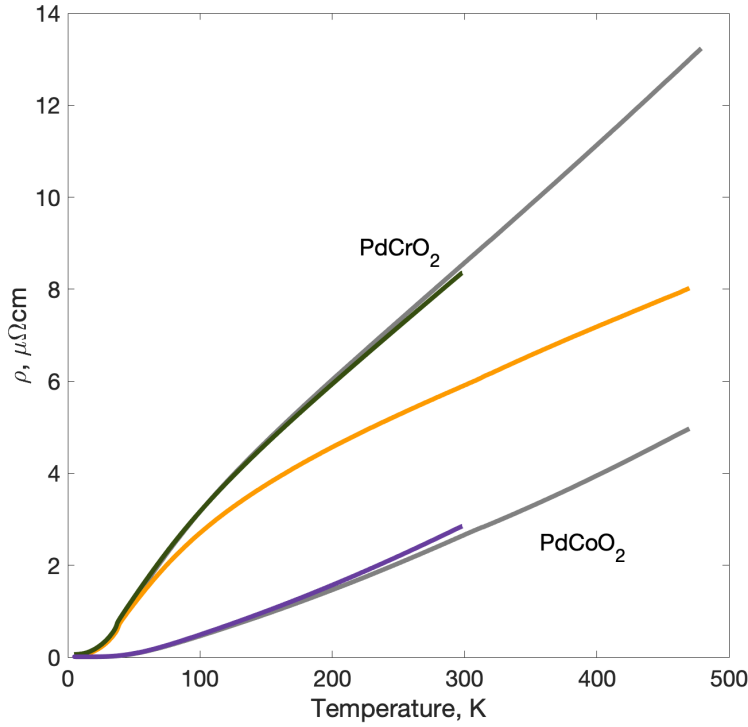


Figure 6: Resistivity of magnetic PdCrO₂ and non-magnetic PdCoO₂ from the microstructures studied in this work (green and purple lines) and the bulk crystals studied in ref. [24] (grey lines), and the difference between the two (orange).

The results presented in Figs. 3 and 6 suggest a Matthiessen's rule approach to understanding the resistivity of PdCrO₂, namely that $\frac{1}{\tau} = \frac{1}{\tau_{imp}} + \frac{1}{\tau_{ph}} + \frac{1}{\tau_{mag}}$ where $\frac{1}{\tau_{imp}}$, $\frac{1}{\tau_{ph}}$ and $\frac{1}{\tau_{mag}}$ are the scattering rates of electrons from impurities, phonons and magnetic excitations respectively. If this is true, it raises a number of interesting questions. In the simplest decomposition into

the different scattering rates, it suggests that although neither $\frac{1}{\tau_{ph}}$ nor $\frac{1}{\tau_{mag}}$ vary as $\frac{k_B T}{\hbar}$ (in either temperature dependence or magnitude), their sum does. This would be remarkable natural fine tuning, unless it resulted from the existence of a scattering rate bound of some kind. As proven by the data in Fig. 3, this bound would have to be not on the absolute value of $\frac{1}{\tau}$ but on its temperature-dependent part. However, a further issue then needs to be considered. The Lorenz ratio data shown in Fig. 4 b) indicate that the scattering is quasi-elastic above approximately 100 K. In the electron-phonon problem this is known to lead to T -linear resistivity because although the scattering at high temperatures is nearly elastic, the cross-section depends linearly on temperature. Crudely, raising the temperature increases the vibration amplitude of the ions, increasing the scattering cross-section [14]. This can also be viewed as the number of excited phonons in each mode being proportional to temperature. For a spin system, however, this is not the case. Once the spins are maximally disordered, their scattering cross-section saturates to some temperature-independent value. The magnetic entropy data shown in Fig. 5 suggest that this is the case by 500 K in PdCrO_2 , yet the magnetic contribution to the resistivity deduced as the blue curve in Fig. 6 continues to rise. This suggests that something more subtle is going on in PdCrO_2 .

We believe that the origin of the observed behaviour may lie in the unusual spectral function reported in ref. [27].

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